

EXPERIMENT - 10

Title

Develop a Job Registration Form and Validate it using AngularJS

Aim

To develop a **Job Registration Form** using AngularJS and implement **form validation** using built-in AngularJS directives.

Prerequisites

- Basic knowledge of HTML & CSS
 - Understanding of AngularJS
 - Knowledge of form validation concepts
-

Software Requirements

- OS: Windows / Linux / macOS
 - Browser: Google Chrome
 - Code Editor: VS Code
 - AngularJS CDN
-

Theory

AngularJS provides powerful form validation using:

- `ng-model`
- `ng-required`
- `ng-minlength`
- `ng-pattern`
- `$valid` and `$invalid`

AngularJS automatically tracks:

- Form state
 - Field validity
 - User interactions
-

Algorithm

1. Include AngularJS CDN.
 2. Initialize AngularJS application using `ng-app`.
 3. Create module and controller.
 4. Design Job Registration Form.
 5. Apply validation directives.
 6. Display validation messages.
 7. Submit only if form is valid.
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Project Structure

None

`fsd_exp10/`

|— index.html

Source Code

index.html

HTML

```
<!DOCTYPE html>
<html ng-app="jobApp">
<head>
  <title>Job Registration Form</title>

  <!-- AngularJS CDN -->

  <script
src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.m
in.js"></script>

  <style>
    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana,
sans-serif;
      background: linear-gradient(135deg, #667eea 0%, #764ba2
100%);
      margin: 0;
      padding: 20px;
      min-height: 100vh;
    }

    .error {
```

```
        color: #d32f2f;
        font-size: 12px;
        margin-top: 5px;
        display: block;
    }

    .success {
        color: #388e3c;
        font-weight: bold;
        background-color: #c8e6c9;
        padding: 15px;
        border-radius: 4px;
        margin-top: 20px;
        text-align: center;
    }

    input, select, textarea {
        padding: 10px;
        margin: 8px 0;
        border: 1px solid #ddd;
        border-radius: 4px;
        font-size: 14px;
        width: 100%;
        box-sizing: border-box;
        transition: border-color 0.3s;
    }

    input:focus, select:focus, textarea:focus {
        outline: none;
        border-color: #667eea;
        box-shadow: 0 0 5px rgba(102, 126, 234, 0.3);
    }
}
```

```
.container {
  width: 600px;
  margin: auto;
  background: white;
  padding: 30px;
  border-radius: 8px;
  box-shadow: 0 4px 20px rgba(0, 0, 0, 0.1);
}

h2 {
  color: #333;
  text-align: center;
  margin-bottom: 30px;
  font-size: 28px;
}

.form-group {
  margin-bottom: 20px;
}

label {
  display: block;
  margin-bottom: 5px;
  color: #555;
  font-weight: 500;
  font-size: 14px;
}

.radio-group, .checkbox-group {
  margin: 8px 0;
}
```

```
.radio-group input[type="radio"],
.checkbox-group input[type="checkbox"] {
    width: auto;
    margin-right: 8px;
    cursor: pointer;
}

.radio-group label,
.checkbox-group label {
    display: inline-block;
    margin-right: 20px;
    margin-bottom: 0;
    font-weight: normal;
    cursor: pointer;
}

button {
    background: linear-gradient(135deg, #667eea 0%, #764ba2
100%);
    color: white;
    padding: 12px 30px;
    border: none;
    border-radius: 4px;
    font-size: 16px;
    cursor: pointer;
    width: 100%;
    font-weight: bold;
    transition: transform 0.2s, box-shadow 0.2s;
    margin-top: 10px;
}
```

```
button:hover:not(:disabled) {
    transform: translateY(-2px);
    box-shadow: 0 4px 12px rgba(102, 126, 234, 0.4);
}

button:disabled {
    opacity: 0.5;
    cursor: not-allowed;
}

.form-row {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 20px;
}

@media (max-width: 768px) {
    .container {
        width: 100%;
        padding: 20px;
    }

    .form-row {
        grid-template-columns: 1fr;
    }
}

</style>
</head>

<body ng-controller="jobController">

<div class="container">
```

```

<h2>Job Registration Form</h2>

<form name="jobForm" novalidate ng-submit="submitForm()">

  <!-- Full Name -->
  <div class="form-group">
    <label>Full Name: *</label>
    <input type="text" name="fullname"
      ng-model="user.fullname"
      ng-required="true"
      ng-minlength="3"
      placeholder="Enter your full name">
    <div class="error" ng-show="jobForm.fullname.$touched &&
jobForm.fullname.$invalid">
      <span
ng-show="jobForm.fullname.$error.required">Name is required.</span>
      <span
ng-show="jobForm.fullname.$error.minlength">Minimum 3 characters
required.</span>
    </div>
  </div>

  <!-- Email -->
  <div class="form-group">
    <label>Email: *</label>
    <input type="email" name="email"
      ng-model="user.email"
      ng-required="true"
      placeholder="Enter your email">
    <div class="error" ng-show="jobForm.email.$touched &&
jobForm.email.$invalid">

```



```

        <span ng-show="jobForm.email.$error.required">Email
is required.</span>
        <span ng-show="jobForm.email.$error.email">Invalid
email format.</span>
    </div>
</div>

<!-- Phone -->
<div class="form-group">
    <label>Phone: *</label>
    <input type="text" name="phone"
        ng-model="user.phone"
        ng-required="true"
        ng-pattern="/^[0-9]{10}$/"
        placeholder="Enter 10-digit number">
    <div class="error" ng-show="jobForm.phone.$touched &&
jobForm.phone.$invalid">
        <span ng-show="jobForm.phone.$error.required">Phone
is required.</span>
        <span ng-show="jobForm.phone.$error.pattern">Enter
valid 10-digit number.</span>
    </div>
</div>

<!-- Qualification -->
<div class="form-group">
    <label>Qualification: *</label>
    <select name="qualification"
        ng-model="user.qualification"
        ng-required="true">
        <option value="">--Select--</option>
        <option>B.Tech</option>
    </select>
</div>

```

```

        <option>M.Tech</option>
        <option>MCA</option>
        <option>B.Sc</option>
    </select>

    <div class="error"
ng-show="jobForm.qualification.$touched &&
jobForm.qualification.$invalid">
        Qualification is required.
    </div>
</div>

<!-- Gender -->
<div class="form-group">
    <label>Gender: *</label>
    <div class="radio-group">
        <label><input type="radio" name="gender"
ng-model="user.gender" value="Male" ng-required="true"> Male</label>
        <label><input type="radio" name="gender"
ng-model="user.gender" value="Female" ng-required="true">
Female</label>
        <label><input type="radio" name="gender"
ng-model="user.gender" value="Other" ng-required="true">
Other</label>
    </div>
    <div class="error" ng-show="jobForm.gender.$touched &&
jobForm.gender.$invalid">
        Gender is required.
    </div>
</div>

<!-- Experience -->
<div class="form-group">

```

```

        <label>Years of Experience: *</label>
        <input type="number" name="experience"
              ng-model="user.experience"
              ng-required="true"
              min="0"
              max="70"
              placeholder="Enter years of experience">
        <div class="error" ng-show="jobForm.experience.$touched
&& jobForm.experience.$invalid">
<span>
        ng-show="jobForm.experience.$error.required">Experienceis
        required.</span>
<span>
        ng-show="jobForm.experience.$error.number">Enteravalid
        number.</span>
</span>
        </div>
    </div>

<!-- Job Position -->
<div class="form-group">
    <label>Job Position Applied For: *</label>
    <select name="position"
          ng-model="user.position"
          ng-required="true">
        <option value="">--Select Position--</option>
        <option>Software Developer</option>
        <option>Senior Developer</option>
        <option>Project Manager</option>
        <option>QA Engineer</option>
        <option>DevOps Engineer</option>
        <option>UI/UX Designer</option>
    </select>

```

```

        <div class="error" ng-show="jobForm.position.$touched &&
jobForm.position.$invalid">
            Position is required.
        </div>
    </div>

<!-- Location -->
<div class="form-group">
    <label>Preferred Location: *</label>
    <select name="location"
        ng-model="user.location"
        ng-required="true">
        <option value="">--Select Location--</option>
        <option>New York</option>
        <option>Los Angeles</option>
        <option>Chicago</option>
        <option>Houston</option>
        <option>Phoenix</option>
        <option>Remote</option>
    </select>
    <div class="error" ng-show="jobForm.location.$touched &&
jobForm.location.$invalid">
        Location is required.
    </div>
</div>

<!-- Skills -->
<div class="form-group">
    <label>Technical Skills: *</label>
    <div class="checkbox-group">
        <label><input type="checkbox"
ng-model="user.skills.javascript"> JavaScript</label>

```

```

                <label><input type="checkbox"
ng-model="user.skills.python"> Python</label>
                <label><input type="checkbox"
ng-model="user.skills.java"> Java</label>
                <label><input type="checkbox"
ng-model="user.skills.nodejs"> Node.js</label>
            </div>
            <div class="checkbox-group">
                <label><input type="checkbox"
ng-model="user.skills.react"> React</label>
                <label><input type="checkbox"
ng-model="user.skills.angular"> Angular</label>
                <label><input type="checkbox"
ng-model="user.skills.mongodb"> MongoDB</label>
                <label><input type="checkbox"
ng-model="user.skills.sql"> SQL</label>
            </div>
            <div class="error" ng-show="jobForm.$submitted &&
!getSkillsSelected()">
                Please select at least one skill.
            </div>
        </div>

<!-- Expected Salary -->
<div class="form-group">
    <label>Expected Salary (Annual): *</label>
    <input type="number" name="salary"
        ng-model="user.salary"
        ng-required="true"
        min="0"
        placeholder="Enter expected salary">

```

```

        <div class="error" ng-show="jobForm.salary.$touched &&
jobForm.salary.$invalid">
            <span
ng-show="jobForm.salary.$error.required">Salary is required.</span>
            <span ng-show="jobForm.salary.$error.number">Enter a
valid number.</span>
        </div>
    </div>

    <!-- Additional Information -->
    <div class="form-group">
        <label>Additional Information:</label>
        <textarea name="additional"
ng-model="user.additional"
placeholder="Any additional information you'd
like to share..."
rows="4"></textarea>
    </div>

    <button type="submit"
ng-disabled="jobForm.$invalid">Register Now</button>

</form>

<br>

<div class="success" ng-show="successMessage">
    {{successMessage}}
</div>
</div>

<script>

```

```

var app = angular.module("jobApp", []);

app.controller("jobController", function($scope) {

    $scope.successMessage = "";
    $scope.user = {
        skills: {}
    };

    $scope.getSkillsSelected = function() {
        var skillsArray = Object.keys($scope.user.skills || {});
        return skillsArray.some(function(skill) {
            return $scope.user.skills[skill] === true;
        });
    };

    $scope.submitForm = function() {
        if ($scope.jobForm.$valid && $scope.getSkillsSelected())
        {
            var registrationData = {
                fullname: $scope.user.fullname,
                email: $scope.user.email,
                phone: $scope.user.phone,
                qualification: $scope.user.qualification,
                gender: $scope.user.gender,
                experience: $scope.user.experience,
                position: $scope.user.position,
                location: $scope.user.location,
                skills:
Object.keys($scope.user.skills).filter(function(skill) {
                    return $scope.user.skills[skill] === true;
                }),
            };
        }
    };

```

```

        salary: $scope.user.salary,
        additional: $scope.user.additional
    };

    console.log("Registration Data:", registrationData);
    $scope.successMessage = "Job Registration
Successful! We will contact you soon.";

    // Reset form after 3 seconds
    setTimeout(function() {
        $scope.$apply(function() {
            $scope.jobForm.$setPristine();
            $scope.jobForm.$setUntouched();
            $scope.user = {
                skills: {}
            };
            $scope.successMessage = "";
        });
    }, 3000);
} else if (!$scope.getSkillsSelected()) {
    console.log("Please select at least one skill");
}

};

});
</script>

</body>
</html>

```

Execution Steps

1. Save file as `index.html`
 2. Open in browser
 3. Try submitting empty form
 4. Enter invalid email / phone
 5. Submit valid data
-

Validation Features Demonstrated

Field	Validation Used
Name	Required + Min Length
Email	Required + Email Format
Phone	Required + Pattern (10 digits)
Qualification	Required Dropdown
Submit Button	Disabled if form invalid

Sample Output

- ✓ If invalid → Red error messages
 - ✓ If valid → "Job Registration Successful!" (Green message)
-

Expected Result

The Job Registration form successfully validates all user inputs using AngularJS validation directives.

Result

Thus, the AngularJS Job Registration form with validation was successfully developed and executed.

Viva Voce Questions

1. What is `ng-model`?
 2. What is `novalidate`?
 3. What is `$touched` in AngularJS?
 4. What is `$valid` and `$invalid`?
 5. How does AngularJS validate email automatically?
 6. What is `ng-pattern`?
-

Conclusion

This experiment demonstrates how AngularJS provides powerful and simple form validation mechanisms for building dynamic web applications.